

Phase 6 – Performance Testing

Introduction

This phase validates the functionality, reliability, and performance of the Personalized Networking Assistant through unit testing, API testing, and manual verification.

Testing Strategy

The application was tested using pytest for unit testing, FastAPI TestClient for API testing, and manual testing through the Streamlit interface. External services such as Wikipedia API were mocked where appropriate to ensure reliable offline tests.

Types of Testing

- Unit Testing
- Integration Testing
- API Testing
- Functional Testing
- Manual UI Testing
- Performance Verification

Performance Results

The application successfully generated personalized conversation starters, extracted event themes, verified factual information, stored history, and recorded feedback. Modular services and fallback mechanisms ensured stable operation even when AI models or external APIs were unavailable.

Observations

The application demonstrated reliable behavior with valid inputs, graceful error handling for invalid requests, and consistent API responses. The modular architecture simplified debugging and future enhancements.

Test Case	Expected Result	Status
Generate Conversation Starters	Relevant AI-generated starters displayed	Pass
Theme Extraction	Correct event themes identified	Pass
Fact Verification	Wikipedia summary returned	Pass
History Storage	Interactions saved successfully	Pass
Feedback Logging	User ratings stored correctly	Pass
Invalid Input	Validation error returned	Pass